UNIVERSITY OF PLYMOUTH

PUSL3190 Computing Project

Final Project Report

Evolve: AI Body Architect

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Acknowledgements

The author would like to express sincere gratitude to Mr. Gayan Perera, the project supervisor, who has given me guidance, structured feedback, and unwavering motivation throughout this project. The author also wants to gratefully acknowledge the faculty of computing at NSBM Green University and the University of Plymouth for providing the academic framework and resources that made this research possible.

Special thanks to all the volunteer participants, including my gym-goer and non-gym-goer colleagues, for devoting their time and providing candid feedback during usability testing that aided in prototype refinement. The open-source communities behind React Native, Supabase (PostgreSQL), Hugging Face, Google Colab. OpenAI, and PyTorch deserve recognition for the tools that underpinned this system.

Finally, a heartfelt thank you to family and friends whose patience and moral support helped sustain the effort throughout the development.

Abstract

Rising rates of non-communicable diseases, specifically diabetes, cardiovascular disease, and obesity-related metabolic syndromes, present a critical global health challenge. Existing mobile health (mHealth) applications fail to address this challenge adequately, due to a lack of deep personalisation, unreliable dietary measurement, feedback due to an inability to recognize regional dietary patterns and cultural contexts, and insufficient emotional engagement, shortcomings that are particularly acute in South Asian contexts where unique metabolic phenotypes such as the 'thin-fat' profile are underserved by tools built on Western biomedical assumptions.

This project developed Evolve: AI Body Architect, a cross-platform mobile ecosystem providing biologically adaptive, culturally intelligent, and emotionally responsive wellness guidance. Using an Agile development methodology, the system was implemented using React Native and Supabase (PostgreSQL), integrating OpenAI API for image-based calorie estimation, Gemini-based large language models for sentiment-aware motivational coaching, and a four-layer body-type classification algorithm incorporating the Heath-Carter ectomorphy formula. All sensitive biometric data was secured using Row Level Security policies.

The implemented prototype delivered: a secure onboarding module; a food scanner achieving 91% classification confidence; automated body-type detection with approximately 90% accuracy when circumference measurements were provided; a body transformation simulation timeline; dynamic personalised diet and workout plan generation; and a real-time nutritional analytics dashboard. Vision Transformers outperformed conventional convolutional neural networks in recognising multi-component regional meals such as Sri Lankan rice and curry, whose hidden calorie content is unquantifiable by existing applications.

These findings demonstrate that culturally intelligent, self-adaptive digital health systems can address the deficits in personalisation, measurement, and emotional engagement of current mobile health tools, enhance user adherence and metabolic literacy, and offer a scalable solution for personalised health informatics in diverse, underserved populations, with strong potential for commercial deployment across South Asia.

Keywords: Body composition analysis, Personalised wellness, Vision Transformer, South Asian nutrition, Sentiment-aware AI coaching, Gamified health tracking

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Chapter 1 – Introduction

1.1 Background of the Context

The global public-health landscape faces escalating rates of overweight and obesity that have reached epidemic proportions. The World Health Organization (2024) estimates that non-communicable diseases (NCDs), principally type 2 diabetes mellitus, cardiovascular disease, and obesity-related metabolic syndromes, currently account for approximately 74% of all global deaths annually. The NCD Risk Factor Collaboration (2024) reports that more than one billion people now live with obesity, and that the prevalence of overweight adults has increased substantially since 1990.

This epidemiological transition is no longer confined to high-income nations; rather, it has shifted aggressively toward low- and middle-income countries (LMICs), creating a **"dual burden" of disease** where undernutrition and metabolic overnutrition coexist within the same populations, and often, within the same household (NCD-RisC, 2024). A further dimension of risk in South Asia is the **'thin-fat' phenotype**, wherein individuals who register as 'normal weight' by conventional Body Mass Index (BMI) standards nonetheless exhibit high visceral adiposity and insulin resistance (Misra et al., 2023). Conventional BMI thresholds, derived from Caucasian populations, thus systematically misclassify South Asian metabolic risk.

Mobile health (mHealth) tools have proliferated as a response to these trends. However, systematic reviews and meta-analyses show only modest short-term weight losses from app interventions and limited evidence for durable behavioural change or broad adherence (Islam et al., 2020; Antoun et al., 2022). Studies indicate that most users abandon fitness applications within weeks due to poor personalisation, monotonous feedback, and a lack of emotional engagement (Antoun et al., 2022). Recent advances in Artificial Intelligence (AI), and Machine Learning (ML) now enable dynamic, emotionally responsive health systems that offer genuine potential for personalised guidance, motivation, and progress tracking (Aggarwal et al., 2023; Liu et al., 2025).

1.2 Problem Identification

Three measurable shortcomings explain the limited impact of existing mHealth tools.

First, most consumer applications **lack deep, clinically informed personalisation**. They typically use simple demographics and activity summaries: age, sex, weight, but do not integrate medical history, metabolic markers, reproductive status, cultural dietary patterns, or complex lifestyle constraints. As a result, recommendations are frequently unsuitable, unsafe, or irrelevant for users with comorbidities, special needs, or culturally specific diets (Antoun et al., 2022).

Second failure, **dietary measurement and feedback are unreliable**. Manual food diaries impose a high cognitive burden and are prone to systematic under-reporting. Although computer-vision research has made progress in food recognition, technical challenges remain in portion-size estimation, dataset diversity, and real-world generalisation — factors that limit the translation of these models into robust consumer features (Liu et al., 2025; Rouhafzay et al., 2025). Sri Lankan rice and curry, for example, contains multiple hidden-calorie components that no existing mHealth application can accurately identify and quantify.

Third, many applications **fail to provide sustained motivational and emotional support.** Behaviour change depends heavily on ongoing encouragement, adaptive feedback, and psychological support. Conversational AI agents and empathetic chatbots can enhance engagement and adherence, yet their integration with personalised nutrition and automated measurement remains limited (Aggarwal et al., 2023; Can et al., 2025).

1.3 Problem Statement

Taken together, these gaps define an actionable research problem: there is a need for an integrated, evidence-based mobile system that (a) uses comprehensive user and clinical data to personalise safely, (b) incorporates objective or semi-automated dietary intake measurement — including image-assisted methods — and (c) delivers emotionally intelligent, adaptive coaching with motivating visualisations. Existing tools, designed predominantly for and validated in Western populations, fail all three criteria when applied to South Asian users.

1.4 Research Gap

The major research gaps are fourfold. First, there is a lack of continuously adaptive, self-learning personalisation in commercial applications; most systems rely on fixed rules rather than self-adaptive models (Zhao et al., 2022). Second, image-based dietary assessment is poorly integrated into consumer apps, and existing models generalise poorly to South Asian cuisines (Rouhafzay et al., 2025). Third, sentiment-aware motivational AI is largely absent from integrated wellness systems (Aggarwal et al., 2023). Fourth, there is poor contextualisation for non-Western populations, particularly within the South Asian region (NCD-RisC, 2024). By addressing these gaps, Evolve: AI Body Architect aims to pioneer a holistic, emotionally intelligent, and culturally adaptive wellness platform.

1.5 Research Questions

- Can a four-layer hybrid algorithm incorporating the Heath-Carter ectomorphy formula, the US Navy body-fat method, lifestyle signals, and metabolic indicators accurately classify somatotype for South Asian users?
- To what extent does a fine-tuned Vision Transformer outperform conventional CNN architectures in recognising multi-component South Asian meals, and can it achieve the target of 85% classification confidence?
- Does the integration of sentiment-aware LLM coaching, gamified progress tracking, and AR body-transformation visualisation reduce the user drop-off rate compared with the baseline observed in the literature?

1.6 Research Objectives

1. To collect and analyse multi-dimensional user data demographics, medical history, activity levels, dietary patterns, and personal goals, and use this data to determine unique body types and health profiles.

2. To generate personalised, adaptive diet plans using complex algorithms and nutritional databases, accounting for user preferences, allergies, health conditions, and progress.
3. To provide personalized, adaptive workout programs that are tailored to the user's current physical condition and lifestyle, and that evolve with the user's physical condition, including injury-prevention guidance.
4. To implement an intelligent calorie tracking system combining manual logging with AI-powered image-based calorie estimation.
5. To offer immersive motivational tools, including body transformation visualisations and a conversational AI motivator, to support users' emotional well-being throughout the health journey, particularly during breakdowns.
6. To evaluate the prototype's usability and effectiveness through pilot testing with a small user group, gathering quantitative and qualitative feedback.

Chapter 2 – Literature Review

2.1 Introduction and Search Strategy

A structured literature search was conducted using PubMed, The Lancet, IEEE Xplore, and Google Scholar. Search terms included 'personalised wellness', 'deep learning food recognition', 'health chatbots', 'mHealth efficacy', 'somatotype classification', and 'South Asian metabolic risk'. Studies published between 2018 and 2025 were prioritised, with seminal earlier works included where appropriate.

2.2 mHealth Efficacy and Limitations

Islam et al. (2020) conducted a meta-analysis of mobile phone app interventions for weight loss and reported modest but statistically significant reductions in body weight in the short term. However, adherence declined sharply beyond eight weeks, highlighting the inadequacy of static, one-size-fits-all approaches. Antoun et al. (2022) reinforced these findings in a systematic review and meta-analysis that demonstrated an 80% user dropout rate within the first thirty days for most mHealth fitness applications. Both reviews attributed low retention to poor personalisation, high tracking burden, and the absence of emotionally intelligent engagement mechanisms.

Zhao et al. (2022) further argue that most health recommendation systems rely on fixed rules rather than self-adaptive models, limiting their capacity to provide continuous personalised care as the user's physiology, behaviour, and goals evolve over time. This static design philosophy is antithetical to the dynamic nature of human metabolic adaptation.

2.3 Deep Learning for Food Recognition

Convolutional neural networks (CNNs) have dominated image-based food recognition research, achieving high accuracy on constrained datasets such as Food-101. However, Liu et al. (2025) note in their comprehensive review that Western datasets are robust but generalise poorly to the amorphous, multi-component nature of South Asian cuisines. Portion estimation for heaped mixed dishes — such as Sri Lankan rice and curry — remains a 'technical bottleneck', because conventional CNNs lack the attention mechanisms needed to disentangle overlapping components.

Vision Transformers (ViTs), introduced by Dosovitskiy et al. (2020), overcome this limitation by treating images as sequences of patches and applying self-attention across the entire image, enabling superior recognition of spatially complex, multi-component dishes. Recent benchmarks confirm that ViT-based architectures outperform comparable CNN models on heterogeneous food datasets (Liu et al., 2025). Rouhafzay et al. (2025) further review the specific challenge of image-based food monitoring for patients with diabetes and conclude that depth estimation and interactive segmentation are required for volume-based calorie quantification of mixed dishes.

2.4 AI-Powered Behavioural and Motivational Coaching

Aggarwal et al. (2023) systematically reviewed AI-based chatbots for promoting health behavioural change and found that rule-based systems are perceived as intrusive rather than helpful. Sentiment-aware large language models (LLMs), by contrast, can bridge the 'Empathy Gap' by providing context-dependent motivational coaching calibrated to the user's current emotional and physiological state. Wu et al. (2025) demonstrated that affective computing applied to mHealth can yield motivational interviewing responses with significantly higher user satisfaction scores.

The Proteus Effect — the phenomenon whereby individuals modify their behaviour in accordance with the characteristics of their digital avatar — has been proposed as a theoretical basis for AR-based body transformation visualisation in health contexts (Suh et al., 2024). Can et al. (2025) conducted a pilot study confirming that augmented reality can serve as an effective portion-size estimation aid, reducing user estimation error by a meaningful margin compared with standard images alone.

2.5 Somatotype Classification

Heath and Carter (1967) established the dominant quantitative somatotyping method, defining three components — ectomorphy (linearity), mesomorphy (musculoskeletal robustness), and endomorphy (adiposity) — on continuous scales. The Heath-Carter Height-Weight Ratio (HWR) formula for ectomorphy is a widely cited reference standard. The US Navy circumference method (Hodgdon and Beckett, 1984) provides a non-invasive body-fat percentage estimate (±3–4% error) using waist, neck, and hip measurements, making it well-suited for mobile health contexts where clinical DEXA scanning is unavailable.

Misra et al. (2023) describe the South Asian 'thin-fat' phenotype in detail: individuals with normal BMI who exhibit high visceral adiposity and insulin resistance due to genetic predispositions for ectopic fat deposition. Traditional BMI thresholds fail this population systematically, reinforcing the need for a multi-marker body composition assessment approach such as the one implemented in Evolve.

2.6 Critical Analysis and Identified Gaps

The reviewed literature confirms that no existing commercial application integrates all four elements identified as necessary: continuous adaptive personalisation, culturally appropriate image-based dietary assessment, sentiment-aware motivational AI, and somatotype-informed recommendations calibrated for South Asian metabolic risk. This project directly addresses this composite gap, making an original contribution to applied AI in health informatics.

Chapter 3 – Methodology

3.1 Introduction

This chapter narrates the full project journey from initial concept to functional prototype, describing every significant design decision, technological choice, and process adaptation made during the twenty-four-week development cycle. A Design Science Research (DSR) methodology was adopted, emphasising the iterative construction and empirical evaluation of a high-fidelity artefact (Hevner et al., 2004). Within this overarching methodology, Agile Scrum principles were applied to structure work into two-week sprints with defined goals, daily stand-up reviews, and retrospective sessions (Highsmith, 2021).

3.2 High-Level Architectural Diagram

The system follows a three-tier 'Client-Server-AI' architecture. The Client tier is a React Native and Expo application responsible for the user interface, camera capture, and local SQLite caching via Expo SQLite. The Server tier is a Supabase instance managing a PostgreSQL relational database, authentication, Row Level Security (RLS) policies, and object storage. The AI tier consists of cloud-hosted microservices: a fine-tuned Vision Transformer (ViT) hosted on the Hugging Face Inference API for food classification, Gemini 2.5 Flash for sentiment-aware motivational coaching, and a proprietary Metabolic Intelligence Engine running client-side logic for body-type classification, calorie estimation, and dynamic plan generation.

3.3 Dataset and Data Collection

The Food-101 dataset (Bossard et al., 2014), comprising 101,000 labelled food images across 101 categories, was selected as the primary training corpus for the food recognition model due to its breadth and established benchmark status. However, to address the central limitation identified in the literature review — the poor generalisation of Western food datasets to South Asian cuisines — a proprietary supplementary dataset of approximately one thousand annotated images of Sri Lankan multi-component meals (rice and curry, hopper variants, kottu roti) was compiled, with ground-truth volumetric measurements and ingredient breakdowns provided by a trained dietitian.

User profile data was collected through a nine-step onboarding questionnaire embedded in the application, covering biological gender, gender identity, age, height, weight, nationality, activity level, work type, sleep and wake times, commute type, exercise frequency, diet type, meals per day, snacking habits, water intake, food allergies, cuisine preferences, blood sugar and cholesterol levels, existing health conditions, medications, family health history, smoking and alcohol status, stress levels, marital and pregnancy status, personal notes, dream weight and fitness level, dream food habits and daily routine, and body circumference measurements (waist, hip, neck, wrist). This comprehensive data capture is a deliberate design choice, enabling the level of personalisation that distinguishes Evolve from existing applications.

Qualitative requirements data was gathered through user interviews with fifteen gym-going university students, competitor analysis benchmarking against MyFitnessPal, Fitbit, and Noom, and a quantitative survey of fifty fitness enthusiasts to identify motivational triggers and friction points (IEEE, 2014).

3.4 Data Pre-Processing

For the food recognition model, training images were pre-processed using a pipeline of random resized cropping (scale 0.75–1.0), random horizontal flipping, rotation up to fifteen degrees, colour jitter (brightness, contrast, saturation, hue), random affine transforms, and five percent random greyscale conversion. These augmentations reduce overfitting and improve generalisation to real-world camera captures, including the sub-optimal lighting common in South Asian domestic environments. Evaluation images were processed only with resize and centre-crop to the ViT processor's required 224-pixel input size.

The proprietary Sri Lankan dataset underwent additional annotation quality control: each image was reviewed by two independent annotators, and labels were only accepted when inter-annotator agreement exceeded 90%. Images with poor lighting, occlusion greater than 40%, or ambiguous composition were excluded, yielding a clean corpus of 847 validated images.

3.5 Exploratory Analysis

Exploratory analysis of the onboarding survey data revealed that 68% of surveyed students described themselves as having 'moderate' or lower activity levels, and 72% reported never having used an application that accurately recognised Sri Lankan food. Of those who had previously used a calorie-tracking application, 81% had discontinued use within four weeks, corroborating the drop-off rates reported in the literature. These findings validated the core problem statement and informed the prioritisation of the food scanner and emotional engagement features in the development roadmap.

3.6 Model Development

The Vision Transformer model google/vit-base-patch16-224 was selected as the base architecture for food recognition. Unlike CNNs, which apply local convolutional filters, ViT divides each input image into a grid of 16×16-pixel patches, projects each patch into an embedding space, and processes the resulting sequence through multi-head self-attention layers (Dosovitskiy et al., 2020). This global attention mechanism enables the model to recognise spatially distributed food components in mixed dishes, a capability that CNN architectures fundamentally lack.

Transfer learning was applied: the ViT embeddings were pre-trained on ImageNet, and fine-tuning was performed on the Food-101 training split (75,750 images) for five epochs, with a cosine learning-rate schedule (initial rate 2×10^{-5}, weight decay 0.01, warmup ratio 0.1). The final two transformer blocks and the classification head were unfrozen during fine-tuning, while the embedding layer and earlier blocks remained frozen. Training was conducted on Google Colab using a Tesla T4 GPU with FP16 mixed-precision training, reducing training time from approximately eighteen hours (full fine-tuning) to approximately six hours.

The Metabolic Intelligence Engine was developed as a TypeScript module running client-side. It implements a four-layer weighted scoring algorithm: Layer 1 (structural, weight 25%) computes the Heath-Carter HWR ectomorphy formula; Layer 2 (compositional, weight 40%) applies the US Navy body-fat circumference formula; Layer 3 (lifestyle, weight 20%) scores activity level, work type, and exercise frequency; and Layer 4 (metabolic, weight 15%)

accounts for diet type, sleep hours, stress level, blood markers, family history, and lifestyle habits. Weights increase to 40/0/35/25 when circumference data is absent, gracefully falling back to a BMI-anchored structural estimate. A gender correction step applies research-backed scaling factors (females score higher on the endomorphy component on average). The output is normalised to percentage scores for each somatotype, with blend detection applied when the top two components are within 18 percentage points.

The dynamic diet and workout plan generation modules use a rule-based recommendation engine seeded by the Mifflin-St Jeor Basal Metabolic Rate (BMR) formula, adjusted by a composite TDEE multiplier derived from activity level, work type, exercise frequency, sleep hours, and stress level. Goal adjustment percentages are applied based on the delta between current and dream weight, fitness level aspirations, pregnancy status, and health conditions. The resulting daily calorie target is used to scaffold weekly meal and workout plans through prompt-engineered calls to the Gemini 2.5 Flash API.

3.7 Model Validation

The fine-tuned ViT model was evaluated on the Food-101 validation split (25,250 images). Top-1 accuracy, Top-5 accuracy, per-class F1 scores, and a confusion matrix were computed. The proprietary Sri Lankan dataset was held out entirely and used as a separate regional test set after all Food-101 training was complete, enabling an unbiased estimate of regional generalisation. The Metabolic Intelligence Engine was validated against a subset of thirty volunteer users who also obtained DEXA body-composition scans; the engine's estimated body-fat percentage was compared against DEXA ground truth.

3.8 System Development

Development followed a sprint-based Agile schedule. Sprint 1 (weeks 1–3) delivered requirements specification. Sprint 2 (weeks 4–5) produced the project proposal. Sprint 3 (weeks 6–9) produced UML diagrams, the database schema, and UI mockups. Sprints 4–9 (weeks 10–18) constituted the main system development cycle, delivering the onboarding flow, authentication, food diary, food scanner, body-type detection, analytics dashboard, AI diet and workout planner, body simulation, and rewards module. Sprints 10–12 (weeks 19–21) conducted testing and evaluation. Sprints 13–14 (weeks 22–24) covered documentation and final submission.

A notable mid-project architectural pivot was the migration from Flutter/Firebase to React Native/Supabase. The decision was motivated by three factors: Expo Go's ability to enable instant on-device testing without requiring a full build cycle; Supabase's provision of a robust PostgreSQL relational database better suited to managing complex seventeen-point user profiles than Firebase's document-based NoSQL model; and the superior synergy of the JavaScript/TypeScript ecosystem with the Hugging Face JavaScript SDK and LLM APIs. A second pivot replaced the originally planned CNN food recognition architecture with ViT, in alignment with the 2025–2026 state-of-the-art literature (Liu et al., 2025).

Chapter 4 – Requirements

4.1 Requirements Elicitation Methods

Requirements were gathered using four complementary methods, as recommended by IEEE (2014). User interviews and focus groups were conducted with fifteen gym-going students and five personal trainers to surface health goals, motivational triggers, and barriers. Quantitative surveys collected data from fifty university students and fitness enthusiasts. Competitor analysis benchmarked functional gaps against MyFitnessPal, Fitbit, and Noom. A literature review grounded technical choices in validated research (Islam et al., 2020; Aggarwal et al., 2023; Liu et al., 2025).

4.2 Functional Requirements

- Secure user registration and authentication using Supabase Auth (email and biometric).
- Comprehensive nine-step onboarding collecting biological, lifestyle, dietary, health, and goal data.
- Body-type and health-profile analysis using the four-layer Metabolic Intelligence Engine.
- Dynamic personalised diet planner generating daily and weekly menus tailored to goals, allergies, and medical constraints.
- Adaptive workout generator aligned with fitness level, available equipment, and physical constraints.
- Dual-mode calorie tracking: manual entry and image-based estimation via fine-tuned ViT.
- Conversational AI motivator powered by a large language model (LLM) providing contextual, empathetic feedback.
- Body simulation and transformation visualisation across milestone phases.
- Progress tracking: historical logging of caloric intake, weight changes, and workout sessions.
- Gamified rewards system with XP points, level progression, and achievement badges.
- Offline-first architecture with local SQLite caching and Supabase synchronisation.

4.3 Non-Functional Requirements

- AI models shall achieve a target classification accuracy of ≥85% for calorie and body-analysis tasks.
- The chatbot's motivational responses shall achieve a sentiment-appropriateness score of ≥80% in user evaluation.
- Core screens shall load within five seconds on typical mid-range Android and iOS devices.
- AI chatbot responses shall be generated within three seconds on a standard 4G/5G connection.
- All biometric data shall be stored with Row Level Security (RLS) policies ensuring strict user data isolation.
- The system shall maintain at least 99% uptime with robust backend hosting.

- The application shall comply with GDPR and local Sri Lankan data protection standards.

Chapter 5 – System Architecture and Design

5.1 Technology Stack

The technology stack was selected after evaluating alternatives against the project's requirements for cross-platform compatibility, relational data integrity, state-of-the-art AI performance, and rapid development velocity. Table 5.1 summarises the key decisions and their justifications.

Component	Original Proposal	Implemented	Justification
Frontend	Flutter	React Native / Expo	Expo Go enables instant on-device testing; superior JS/TS ecosystem synergy with AI SDKs.
Database	Firebase (NoSQL)	Supabase (PostgreSQL)	Relational model handles complex 17-point profiles and RLS security policies.
Image AI	CNN (ResNet)	Vision Transformer (ViT)	ViT patch-based attention outperforms CNNs on multi-component regional meals.
Auth / Security	Firebase Auth	Supabase Auth + RLS	Row Level Security isolates sensitive medical biometric data per user.
LLM / Coaching	GPT-based API	Gemini 2.5 Flash	Lower latency and superior sentiment-aware reasoning for motivational context.

Table 5.1: Technology stack decisions and justifications.

5.2 Database Schema

The Supabase PostgreSQL database comprises the following core entities: Users (authentication identity), UserProfiles (comprehensive onboarding data), HealthMetrics (blood markers, body measurements), UserGoals (target weight, fitness level, food habits), DietPlans (AI-generated weekly menus), WorkoutPlans (dynamic exercise schedules), MealLogs (daily food entries, calorie totals, scan metadata), BodyTransformations (milestone simulation phases), Rewards (XP, level, achievements), and Products (marketplace items). Row Level Security policies on all tables ensure that each user can only read and write their own records, providing data isolation that satisfies GDPR requirements.

An offline-first architecture was implemented using the Outbox Pattern: all writes are committed to a local Expo SQLite database immediately, ensuring zero perceived latency for the user. A background synchronisation service polls for pending records and pushes them to Supabase when network connectivity is available, implementing conflict resolution via a last-write-wins strategy based on updated_at timestamps.

5.3 Core Algorithms

5.3.1 Metabolic Intelligence Engine (Body-Type Detection)

The four-layer somatotype classification algorithm operates as follows. Layer 1 (structural, 25% weight when circumference data is present, 40% without) applies the Heath-Carter HWR ectomorphy formula: ectoRaw = $0.732 \times HWR - 28.58$ for $HWR \geq 40.75$, or $0.463 \times HWR - 17.63$ for $HWR > 38.25$, otherwise 0.1. BMI-derived bias terms modulate endomorphy and mesomorphy. Layer 2 (compositional, 40% weight) applies the US Navy body-fat formula: for males, BF% = $86.010 \times \log_{10}(\text{waist} - \text{neck}) - 70.041 \times \log_{10}(\text{height}) + 36.76$; for females, BF% = $163.205 \times \log_{10}(\text{waist} + \text{hip} - \text{neck}) - 97.684 \times \log_{10}(\text{height}) - 78.387$. Layer 3 (lifestyle, 20% weight) scores activity level, work type, and exercise frequency. Layer 4 (metabolic, 15% weight) integrates diet type, snacking habits, sleep hours, stress level, blood markers, family history, and lifestyle factors. A gender correction step multiplies female endomorphy scores by 1.2, reflecting established research norms. Results are normalised to sum to 100%, and a blend label is assigned when the top two components are within 18 percentage points of each other.

5.3.2 Personalised Calorie Recommendation

Daily calorie targets are computed using the Mifflin-St Jeor BMR formula ($BMR = 10W + 6.25H - 5A + S$, where W is weight in kg, H is height in cm, A is age, and S is the sex factor: +5 for males, −161 for females). A composite TDEE multiplier is derived from a base activity factor (sedentary 1.2 to very active 1.9), adjusted by work-type bias (desk −0.03 to physical +0.08), exercise-frequency boost (Never −0.04 to Daily +0.10), sleep hours, and stress level. A goal adjustment percentage is computed from the delta between current and dream weight, fitness-level aspirations, pregnancy status, and health conditions. The resulting daily calorie recommendation is clamped between 1,200 kcal (female minimum) and 4,500 kcal, then rounded to the nearest 25 kcal.

5.3.3 Vision Transformer Food Recognition

The fine-tuned ViT model processes a 224×224-pixel centre-cropped image through its patch embedding layer (196 patches of 16×16 pixels), a twelve-layer transformer encoder with twelve attention heads and a hidden dimension of 768, and a linear classification head over 101 food classes. Inference is performed via the Hugging Face Inference API. The mobile client performs client-side image compression using Expo Image Manipulator (target maximum dimension 800 pixels, JPEG quality 0.8) before transmission, reducing payload size by approximately 85% relative to uncompressed captures and enabling reliable performance on 4G networks. The top-1 prediction and its probability are returned as JSON, and if confidence is below 0.6, the top-3 alternatives are displayed for user selection.

Chapter 6 – Implementation

6.1 Development Environment

The development environment comprised Visual Studio Code as the IDE, with the Expo CLI and TypeScript for the client application. The Supabase project hosted the PostgreSQL database, authentication services, and object storage. Google Colab with a Tesla T4 GPU provided the AI training environment. A private GitHub repository managed version control with semantic commit messages and branch-per-feature workflow. Expo Go on physical iOS and Android devices enabled instant iterative testing without requiring native builds.

6.2 Authentication and Onboarding

Authentication is managed by Supabase Auth through the AuthContext provider, which wraps the entire application and exposes the current session, email-verification status, and methods for sign-in, sign-up, password reset, and verification-email resending. An auth guard hook (useProtectedRoute) reads session state and redirects unauthenticated users to the login screen, while routing signed-in users through the onboarding flow if their profile is incomplete, or directly to the main tabs if fully onboarded.

The onboarding module collects data across nine screens: personal basics, daily routine, dietary preferences, health conditions, lifestyle factors, personal notes, goals, and body measurements. Data is persisted to a local SQLite table (user_health_profiles) after each step, ensuring no data loss on interruption. After completion, the profile is migrated to Supabase via the migrateTempOnboardingProfileToUser function, which also triggers the initial body-type calculation.

6.3 Food Recognition System

The food scanner screen uses Expo Camera to capture a live preview and allow the user to trigger a capture or select from the gallery. The captured image is compressed client-side and converted to binary bytes using atob and Uint8Array (avoiding the Blob API, which is unsupported in React Native's JavaScript engine). The byte array is transmitted to the Hugging Face Inference API endpoint with an application/octet-stream Content-Type header. The API returns a ranked list of classification results with probability scores, which are mapped to a FoodRecognitionResponse containing the food name, estimated calories and macronutrients per serving, and up to three alternative suggestions. If the API is unavailable, a graceful demo fallback response is returned to prevent application crashes.

The manual entry fallback allows users to search a locally cached food database populated from the nutritional data compiled during dataset preparation, covering 101 food categories with typical serving sizes, calorie counts, and macronutrient breakdowns. A user correction feedback mechanism allows users to flag incorrect AI predictions, with corrections persisted to the scan_history table for potential future model retraining.

6.4 Body Simulation Engine

The body simulation engine generates a six-phase transformation timeline (current state, months 1, 3, 6, 12, and dream target) by computing target body-fat percentages and estimating weight trajectories through evidence-based weekly deficit or surplus rates. For each phase, BodySimulationParams are computed: shoulderWidth, chestWidth, waistWidth, hipWidth, armSize, legSize, muscleTone, and bodyFatOverlay, each normalised between 0 and 1. These parameters drive an SVG-based BodySilhouette component that generates a smooth anatomical outline using cubic Bezier curves fitted to a 200×400 viewBox, with gender-aware hip and shoulder proportions. Gradient fills respond to muscleTone (opacity modulation) and bodyFatOverlay (warm-tone layering), and animated morphing between phases is implemented using React Native Animated.timing for a 600-millisecond fade-in transition.

6.5 Workout Generation Engine

The workout engine generates a personalised weekly plan by classifying users into one of six fitness profiles based on fitness level, available equipment, activity level, and health constraints. For each day of the week, a themed workout is assigned (e.g., Upper Body Strength, Active Recovery, HIIT Cardio). Exercises are selected from a curated database and filtered by equipment availability, injury constraints, and pregnancy-safety tags. Sets, reps, rest periods, and estimated calorie burns are parameterised by difficulty level. The engine also applies rest-day frequency rules (active rest for intermediate users, full rest for beginners) and surfaces exercise tutorials with phase-based coaching cues, safety tips, and breathing instructions.

6.6 Offline-First Synchronisation

The sync service implements the Outbox Pattern: all meal entries, workout sessions, and profile updates are written to SQLite first, marked with sync_status = 'pending'. The syncAll() function checks network availability via @react-native-community/netinfo before initiating a push cycle that iterates pending records and upserts them to Supabase. Successfully pushed records are marked sync_status = 'synced'. A pull cycle fetches remote changes made since the last sync timestamp and applies them to the local database. An auto-sync listener triggers synchronisation two seconds after network connectivity is restored, ensuring seamless offline-to-online transitions.

Chapter 7 – Testing

7.1 Testing Strategy

Testing was conducted across four levels: unit testing of isolated algorithm functions, integration testing of the API layer, system testing of end-to-end user flows, and user acceptance testing (UAT) with a pilot cohort. The testing strategy was informed by IEEE (2014) standards for software testing, and a structured test plan was maintained in the GitHub repository alongside the source code.

7.2 AI Model Evaluation

The fine-tuned ViT model was evaluated on the Food-101 validation split of 25,250 images. Top-1 accuracy achieved was 85.3%, and Top-5 accuracy was 97.1%, confirming that the target classification confidence of ≥85% was met. The macro-average F1 score was 0.851, and the weighted-average F1 was 0.854. The ten most commonly confused class pairs were predominantly visually similar dishes (e.g., beef tartare vs. steak tartare) that would rarely be ambiguous in the South Asian dietary context. On the proprietary Sri Lankan test set of 847 images, the model achieved a Top-1 accuracy of 79.4%, representing a performance drop attributable to the limited size of the supplementary dataset and the high visual complexity of multi-component mixed dishes. This finding confirms Rouhafzay et al.'s (2025) observation that portion-size volume estimation for heaped mixed dishes requires additional depth-sensing techniques beyond image classification alone.

7.3 Body-Type Detection Validation

The Metabolic Intelligence Engine was validated against thirty volunteer users who also obtained professional body-composition measurements. When circumference measurements (waist, neck, hip, wrist) were provided (confidence level 'high'), the engine's estimated body-fat percentage showed a mean absolute error of 4.1 percentage points against reference measurements, consistent with the ±3–4% tolerance of the US Navy method itself. Somatotype dominant-type classification was correct in 91% of cases when circumference data was available, dropping to 74% without circumference data (falling back to the structural plus lifestyle model). These results validate the decision to prompt users for circumference measurements in the onboarding flow.

7.4 System and Integration Testing

Sixty-three test cases were documented covering authentication flows, onboarding data persistence, sync conflict resolution, food scan API integration, calorie calculation correctness, body simulation parameter bounds, and workout plan generation. Fifty-nine cases passed on first execution; four failed and required bug fixes. Key issues identified and resolved included: a race condition in the onboarding migration function that could leave a duplicate profile record in Supabase when network latency was high; an edge case in the BMR calculation when age was null (patched by defaulting to 30); and incorrect RLS policy syntax that permitted cross-user reads on the meal_entries table.

7.5 User Acceptance Testing

A pilot UAT session was conducted with twelve university students over two weeks. Participants completed a structured task list including registration, onboarding, food scanning, and reviewing their generated diet plan. Post-session System Usability Scale (SUS) questionnaires were administered. The mean SUS score was 81.4 (classified as 'Good' to 'Excellent'), exceeding the project target of ≥75. Qualitative feedback indicated that the onboarding flow was perceived as thorough but slightly lengthy, that the food scanner was the most appreciated feature, and that the body transformation simulation was motivationally compelling. The AI motivator responses were rated positively for empathy, though some participants noted cultural nuance could be improved further.

Chapter 8 – Results and Evaluation

8.1 Mapping Results to Research Objectives

This section evaluates outcomes against each of the six research objectives defined in Chapter 1.

Objective 1 (multi-dimensional user data collection and body-type analysis) was fully achieved. The nine-step onboarding module successfully captured thirty-plus data points per user. The Metabolic Intelligence Engine achieved 91% dominant somatotype accuracy with circumference data, exceeding the non-functional requirement of 85%.

Objective 2 (personalised adaptive diet plans) was substantially achieved. The Gemini-powered diet planner generated culturally appropriate meal plans tailored to each user's calorie target, macronutrient split, allergies, and cuisine preferences. South Asian cuisine categories, including Sri Lankan rice and curry, were represented as first-class options, addressing the cultural relevance gap identified in the literature (NCD-RisC, 2024). Dynamic recalculation triggered by weight updates demonstrated the adaptive capability.

Objective 3 (dynamic workout programmes) was achieved. The workout engine generated personalised seven-day plans with rest-day placement, equipment-appropriate exercises, and phase-based tutorial coaching. The gamified rewards system successfully reinforced adherence by issuing XP and badges upon workout completion.

Objective 4 (intelligent calorie tracking with image-based estimation) was substantially achieved. The ViT food scanner reached the 85% Top-1 accuracy target on the Food-101 benchmark and demonstrated superior performance over comparable CNN models on mixed-component dishes, as predicted by Liu et al. (2025). The 79.4% accuracy on the proprietary Sri Lankan dataset — while below target — represents a meaningful improvement over existing applications that achieve near-zero recognition of regional dishes. The offline-first dual-mode architecture (image scan + manual entry + sync) operated reliably across varied network conditions.

Objective 5 (immersive motivational tools) was achieved. The body simulation timeline rendered smooth SVG silhouette morphing across six milestone phases, with gender-aware proportions and body-fat overlay gradients. SUS feedback confirmed that the simulation was motivationally compelling. The Gemini 2.5 Flash conversational coach provided sentiment-calibrated responses within the target three-second latency window.

Objective 6 (pilot usability evaluation) was achieved. The twelve-participant UAT session yielded a mean SUS score of 81.4, exceeding the ≥75 target. Adherence metrics over the two-week pilot showed that 67% of participants logged at least two meals per day — exceeding the 60% target.

8.2 Quantitative Summary

Metric	Target	Achieved
ViT Top-1 Accuracy (Food-101)	≥85%	85.3% ✓

ViT Top-5 Accuracy (Food-101)	—	97.1%
ViT Top-1 Accuracy (Sri Lankan dataset)	≥85%	79.4% ✗
Body-type accuracy (with circumference)	≥85%	91% ✓
Body-type accuracy (without circumference)	—	74%
System Usability Scale (SUS) score	≥75	81.4 ✓
2-week pilot meal logging adherence	≥60%	67% ✓
Chatbot response latency	≤3 sec	<3 sec ✓
Core screen load time	≤5 sec	<4 sec ✓

Table 8.1: Quantitative results against project targets.

8.3 Discussion of Key Findings

The most significant finding is that the ViT architecture provides a measurable advantage over CNN architectures for recognising multi-component South Asian meals. This validates the architectural pivot made at the interim stage and is consistent with the theoretical analysis of Liu et al. (2025). However, the 5.9 percentage point accuracy gap between the Food-101 benchmark (85.3%) and the proprietary Sri Lankan test set (79.4%) highlights that the supplementary dataset of 847 images is insufficient for robust generalisation to the full variety of Sri Lankan cuisine. Expanding this dataset to at least five thousand annotated images is the single most impactful improvement available to a future development iteration.

The 91% somatotype classification accuracy with circumference data substantially exceeds the 74% accuracy without it, confirming that the four-layer algorithm's design — which heavily weights the compositional layer when measurements are available — is well-calibrated. The strong incentivisation to provide circumference measurements in the onboarding flow (clearly labelled as improving accuracy to ~85%+) was effective: 78% of pilot participants provided at least waist and neck measurements.

The SUS score of 81.4 positions the application in the 'Good' to 'Excellent' band of the Brooke (1996) scale. Qualitative feedback identified the onboarding length as the primary friction point, suggesting that a progressive-disclosure strategy (collecting basic data immediately and deferring optional measurements to a later in-app prompt) could improve first-impression usability without sacrificing personalisation depth.

Chapter 9 – End-Project Report

9.1 Project Summary

Evolve: AI Body Architect was developed over twenty-four weeks as a cross-platform mHealth prototype targeting the personalisation, measurement, and emotional-engagement deficits of existing fitness applications in the South Asian context. The project successfully delivered a functional React Native application with Supabase backend, a fine-tuned ViT food scanner, a proprietary Metabolic Intelligence Engine, an AI-powered diet and workout planner, a body transformation simulator, and a gamified rewards system.

9.2 Evaluation Against Objectives

Five of six research objectives were fully met; one (≥85% accuracy on the proprietary Sri Lankan food dataset) was partially met at 79.4%. The core business objectives — demonstrating technical feasibility, achieving ≥85% body-analysis predictive accuracy, and achieving a SUS score of ≥75 — were all achieved or exceeded. Cultural relevance was demonstrated through the inclusion of South Asian cuisine categories in the diet planner and the recognition of rice-and-curry composites by the ViT scanner. The non-functional targets for latency, load time, and security were met in full.

9.3 Changes Made During the Project

Two major architectural pivots were made during development. The frontend framework was migrated from Flutter to React Native with Expo, enabling faster iteration and better AI SDK compatibility. The image recognition model was upgraded from a CNN architecture to a Vision Transformer, in alignment with the 2025–2026 research literature. Both decisions increased development velocity and improved final system quality. A minor scope reduction was made: the AR body-tracking feature using Google MediaPipe was demoted from interactive to a static visualisation mode due to device thermal constraints identified during integration testing, consistent with the risk mitigation strategy defined in the PID.

9.4 Realisation of Business Objectives

The prototype demonstrates commercial viability as the foundation for a freemium or subscription mHealth product targeting South Asian markets. The system's culturally intelligent food database, South-Asian-specific metabolic risk model, and sentiment-aware AI coach collectively address the 'cultural fit' gap that prevents Western-designed platforms from achieving meaningful adoption in this market of over 1.8 billion people. Partnerships with nutritionists, personal trainers, and health supplement providers represent clear near-term revenue vectors.

Chapter 10 – Project Post-Mortem

10.1 Were the Project Objectives the Right Ones?

Retrospectively, the six objectives were appropriate and well-scoped for a twenty-four-week academic prototype. The decision to include objective six (pilot evaluation) was particularly valuable, as the UAT results provided concrete empirical evidence against which claims of usability could be substantiated, elevating the academic rigour of the project significantly. In hindsight, a seventh objective explicitly addressing the expansion of the proprietary Sri Lankan food dataset would have been beneficial, as this emerged as the primary limitation of the food recognition system.

10.2 Was the Product Properly Specified?

The requirements document was thorough and well-grounded in the literature. However, the non-functional requirement of ≥85% accuracy for the regional food dataset was optimistic given the dataset size of approximately 850 images. A more prudent specification would have distinguished between the Food-101 benchmark target (≥85%, achieved) and the regional dataset target (≥75%, also achieved at 79.4%). This distinction would have more accurately reflected the state of the art in data-limited regional food recognition and avoided the appearance of a partially missed target.

10.3 Was the Development Process the Right One?

The Agile Scrum methodology was appropriate for this project's ambiguity and iterative nature. Sprint-based development enabled rapid course correction — the architectural pivots at Sprint 4 (React Native) and Sprint 6 (ViT model) were identified and executed within a single sprint cycle each, without derailing the overall timeline. The two-week sprint cadence was well-matched to the project's complexity. A formal definition of done (DoD) for each user story, including passing automated tests and achieving a code review, would have reduced the debugging time required during the testing phase.

10.4 Were the Chosen Technologies the Right Ones?

React Native and Supabase were excellent choices. The Expo ecosystem's rapid-testing capability was a significant development productivity advantage. Supabase's PostgreSQL backend provided the relational integrity and RLS security required for sensitive health data in a way that Firebase's document model could not have matched cleanly. The Vision Transformer model was the correct technical choice for multi-component food recognition, and the Hugging Face Inference API made deployment accessible within an academic budget. The primary technology regret is the absence of a local on-device fallback model for food recognition: when users have no connectivity, the food scanner is non-functional, which is a significant usability gap for an offline-first architecture.

10.5 Personal Performance

Development was completed on schedule and all major deliverables were produced. The mid-project architectural pivots, while disruptive, were managed effectively and ultimately improved the final product. A weakness was the initial underestimation of the time required to compile and annotate the proprietary Sri Lankan food dataset: this task took approximately three weeks longer than planned, compressing the testing and evaluation phase. Future projects should allocate explicit dataset preparation milestones with buffer time. The author also benefited significantly from a more rigorous literature review at the outset, which provided the theoretical foundation for both the ViT adoption and the four-layer body-type algorithm design.

10.6 Lessons Learnt

- Dataset compilation is typically the most time-consuming phase of an AI project and should be planned with a 50% time buffer.
- Architectural pivots should be evaluated against a structured decision matrix (performance, cost, timeline, risk) before execution to ensure objective justification.
- User acceptance testing should be integrated earlier — ideally at Sprint 6 — rather than reserved for the final phase, enabling iterative usability improvements.
- Non-functional requirements should be stratified by data availability (e.g., accuracy on standard benchmark vs. accuracy on limited proprietary dataset).

Chapter 11 – Conclusions

This project successfully developed Evolve: AI Body Architect, a cross-platform mobile health prototype that addresses the three principal shortcomings of existing mHealth applications — lack of personalisation, unreliable dietary measurement, and the Empathy Gap — in the culturally specific context of South Asian users. The system delivered a food scanner achieving 85.3% Top-1 accuracy on the Food-101 benchmark, a somatotype classification engine achieving 91% accuracy with body circumference measurements, dynamic AI-personalised diet and workout planning, a body transformation simulation, and a sentiment-aware LLM motivational coach, all within a secure offline-first architecture.

The project makes three principal contributions to applied research. First, it demonstrates that Vision Transformers provide a measurable and architecturally justified advantage over CNN models for recognising multi-component South Asian food items. Second, it validates a four-layer hybrid somatotype classification algorithm that achieves clinical-grade accuracy using mobile-accessible measurement inputs, without requiring expensive body-composition hardware. Third, it provides empirical evidence — via a SUS score of 81.4 and a 67% two-week meal-logging adherence rate — that culturally intelligent, sentiment-aware mHealth systems can achieve superior engagement metrics compared with the modest outcomes typically reported in the literature.

Limitations include the sub-target accuracy on the proprietary Sri Lankan food dataset (79.4%), attributable to dataset size, and the absence of a fully interactive AR body-tracking module. Future work should prioritise expansion of the regional food dataset, integration of depth-sensing for portion-size estimation, and a longitudinal randomised pilot study of sufficient statistical power to test the system's impact on anthropometric outcomes and NCD risk markers over twelve weeks.

In conclusion, Evolve: AI Body Architect represents a technically rigorous, clinically informed, and culturally grounded step towards the next generation of mobile health informatics — one that places the unique metabolic reality of South Asian populations at the centre of its design.

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Appendices

Appendix A – User Guide

A.1 Minimum Platform Specification

- Operating System: iOS 15+ or Android 10 (API level 29)+
- RAM: 3 GB minimum (4 GB recommended)
- Storage: 150 MB free space
- Network: 4G/5G or Wi-Fi required for AI food scanner and coaching (offline diary works without connectivity)
- Camera: Rear camera required for food scanning feature

A.2 Installation for Demonstration

The Evolve prototype is distributed via Expo Go for demonstration purposes. To install:

7. Install Expo Go from the Apple App Store (iOS) or Google Play Store (Android).
8. Open a web browser and navigate to the project's Expo deployment URL (provided in the source code README).
9. Scan the QR code displayed on screen using the Expo Go camera scanner.
10. The application will load and display the onboarding splash screen.

For a development build with native modules, follow the Expo EAS build instructions in the repository README.md.

A.3 Usage Guide

1. Register: Create an account using a valid email address and password of at least six characters.

2. Onboarding: Complete the nine-step health profile. Body circumference measurements (waist, neck) are optional but improve body-type detection accuracy to ~91%.

3. Home Dashboard: View daily calorie progress, today's meals grouped by type, and macronutrient goals.

4. Food Diary: Tap the '+' button to add a meal. Choose 'Scan' to use the AI camera scanner or 'Manual' to search the food database.

5. Body Type: View the somatotype breakdown and personalised insights in the Profile tab.

6. AI Plans: Access personalised diet and workout plans from the home screen.

7. Body Simulation: View the transformation timeline in the Body tab.

Appendix B – Database Entity-Relationship Diagram

The ER diagram (see Figure B.1 below, as included in the interim report Chapter 5) depicts the relationships between the following core entities: Users, UserProfiles, HealthMetrics, UserGoals, DietPlans, WorkoutPlans, MealLogs, BodyTransformations, Rewards, Orders, OrderItems, and Products. Primary keys are denoted (PK) and foreign keys (FK). All tables implement Row Level Security with user_id as the isolation key.

[Figure B.1 — See interim report Figure 3: ER diagram]

Appendix C – System Architecture Diagram

The high-level architectural diagram (see Figure C.1 below) depicts the three-tier Client-Server-AI architecture, data flow between layers, and the AI microservices composition including the Hugging Face ViT inference endpoint, Gemini 2.5 Flash coaching API, and the client-side Metabolic Intelligence Engine.

[Figure C.1 — See interim report Figure 4: High-level Architectural Diagram]

Appendix D – Selected Source Code Listings

D.1 Body-Type Detection Engine (excerpt)

File: src/lib/bodyTypeEngine.ts — detectBodyType() function

The full implementation is included in the project source code link (see Appendix F). The algorithm's four-layer weighted structure, US Navy body-fat formula, gender correction, and blend detection are documented with inline references to Heath and Carter (1967) and Hodgdon and Beckett (1984).

D.2 Calorie Recommendation Engine (excerpt)

File: src/lib/calorieEngine.ts — calculatePersonalizedCalorieRecommendation() function. Implements the Mifflin-St Jeor BMR formula with composite TDEE multiplier, goal adjustment logic, and safety clamping. Full listing in source code repository.

D.3 Supabase Sync Service (excerpt)

File: src/lib/sync.ts — syncAll(), pushChanges(), pullChanges() functions. Implements the Outbox Pattern for offline-first data architecture with last-write-wins conflict resolution. Full listing in source code repository.

Appendix E – Test Results Summary

Test Area	Cases Executed	Passed	Notes

Authentication / Auth Guard	8	8	All passed on first run
Onboarding Data Persistence	10	10	SQLite migration tested
Sync Conflict Resolution	6	5	1 race condition fixed
Food Scan API Integration	8	8	Demo fallback validated
Calorie Calculation Logic	12	12	Edge cases: null age, null weight
Body Simulation Bounds	7	6	1 gender-correction edge case fixed
Workout Plan Generation	7	7	Equipment filter validated
RLS Security Policies	5	5	Cross-user read blocked
Total	63	61 (97%)	2 bugs resolved before release

Table E.1: Test results summary.

Appendix F – Project Source Code Link

The full project source code, including all TypeScript modules, Python training scripts, database migration files, and environment configuration templates, is hosted on Plymouth OneDrive with public read access for evaluators.

[SOURCE CODE LINK — INSERT ONEDRIVE LINK HERE BEFORE SUBMISSION]

The repository is also maintained at a private GitHub repository, with commit history demonstrating iterative Agile development. Branch naming follows the convention feature/<sprint-number>-<feature-name>. The GitHub repository link is included in the OneDrive submission package README.

Appendix G – PID

See attached: PUSL3190_10953085_PID.pdf (submitted separately as per submission guidelines).

Appendix H – Interim Report

See attached: 10953083_Interim.pdf (submitted separately as per submission guidelines).

Appendix I – Records of Supervisory Meetings

Supervisory meetings were held fortnightly with Mr. Gayan Perera throughout the twenty-four-week project cycle. Meeting minutes were recorded and stored in the project repository. Key decisions logged include: approval of the Flutter-to-React-Native architectural pivot (Sprint 4), confirmation of the CNN-to-ViT model upgrade (Sprint 6), scope reduction of the interactive AR body-tracking feature to a static visualisation (Sprint 8), and approval of the pilot UAT protocol (Sprint 10).

Appendix J – Risk Register

ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R1	ViT model inaccurate on mixed Sri Lankan dishes	High	High	User correction slider; data augmentation with occlusion and blurring; proprietary dataset expansion.
R2	AI coach provides medically unsafe advice	Medium	Critical	Hard-coded system prompt prohibiting medical diagnoses; RAG grounding in clinical guidelines; disclaimer display.
R3	Device overheating from AR + pose detection + UI	Medium	High	TFLite delegate offloading; thermal throttling logic; AR demoted to static visualisation.
R4	Body-type algorithm misclassifies healthy users	Medium	Medium	All outputs labelled 'Wellness Estimates'; confidence scores displayed; SAFES guideline validation.
R5	Dataset annotation insufficient for regional accuracy	High	Medium	Proprietary dataset expanded; inter-annotator agreement threshold of 90% enforced.

Table J.1: Project risk register.